

Falcon 2.0T crank/cam correlation P0016 due to timing belt tensioner

Document: tsb-21-223

Area: bulletins

1 Condition

Falcon 2.0T vehicles may store diagnostic trouble code P0016 (Crankshaft Position - Camshaft Position Correlation, Bank 1 Sensor A) accompanied by a hard start, rough idle, or reduced power complaint. The malfunction indicator lamp will illuminate and the engine control module may limit timing advance as a protective measure. Confirm the P0016 fault is current and inspect for related correlation codes before disassembly. General engine architecture and sensor locations are described in the Falcon Engine Service Manual .

2 Cause

This condition is caused by wear and tension loss in the automatic timing belt tensioner assembly, part number TB-3322, which allows the belt to skip a tooth and shift the camshaft timing relative to the crankshaft. As the TB-3322 hydraulic damper loses preload, belt slack under cold start torque triggers the P0016 correlation fault. The detailed inspection criteria for the tensioner are covered in the Falcon Engine Service Manual :: Timing System chapter. A comparable failure mode and identification guidance is documented in the Harrier Engine Service Manual :: Timing Sub-System .

3 Repair Procedure

Replace the timing belt tensioner TB-3322 together with the timing belt as a set, following procedure PROC-TIMING-BELT, and verify all timing marks align before rotating the engine by hand two full revolutions. After installation, clear P0016 and confirm correlation is within specification on a road test. The full procedure is provided in the Falcon Engine Service Manual :: Timing Belt and Tensioner Service section. For cross-platform reference, equivalent steps appear in the Harrier Engine Service Manual :: Timing Belt and Tensioner Replacement and belt specifications in the Lumen Engine Service Manual :: Timing Belt Specifications and Identification .